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Research Statement

Overview: Measuring Collective Decisions

Cities increasingly use voting rules to decide which projects to fund from their participatory budgets, and large language models are being tested for summarizing public consultations and for drafting statements that a whole group can agree on. Whether a voting rule or a summary treats the participants fairly can be studied in theory, but it also has to be tested, which requires test cases that resemble real decisions and real data to test on. Computational social choice, the field that studies voting rules with the tools of computer science, long lacked both. Its experiments used synthetic votes, often drawn uniformly at random (known as the impartial culture), with no general method to check how close these were to real elections.

My work has provided both for decisions made with ballots, along three lines of research.

- *A geometry of collective decisions.* I introduced *maps of elections*, which place elections in a common space, using a distance that I proposed, so that similar elections lie close together and one can see which models of synthetic votes resemble real data [C3, J4].
- *Real data and practice.* I founded Pabulib, the largest open repository of participatory budgeting data. It holds over 2,000 elections from 8 countries and has been cited in more than 100 papers. I am also a member of the team that convinced Polish cities to adopt the Method of Equal Shares, a proportional rule for participatory budgets.
- *Toward language.* I now study collective decisions in which text and language models play a role, including simulated deliberation among LLM agents [C28] and the UN General Assembly, where states vote on texts [P2].

As a PhD student, I was the principal investigator of a National Science Centre grant (NCN Preludium) on the *election isomorphism problem*, from which maps of elections grew. My thesis, *Map of Elections* (advisor: Piotr Faliszewski), received the Polish Prime Minister's Award (2025); I also received a START fellowship (2023) and the AAMAS 2023 Best Student Paper Award. I was then a CNRS postdoc at LAMSADE with Jérôme Lang (2024–2026), and I am now a postdoc at the University of Geneva.

Collective decisions increasingly start from text: people write what they think, and AI condenses it into a single summary. At EPFL, I will build the tools to check whom such a summary represents, and whom it leaves out. Examples include AI-written summaries of public consultations and consensus statements. Methods that generate such texts already exist, and some come with guarantees for the texts they produce, but there is little work on checking whom a given summary represents, whatever method produced it. For example, given the comments that residents submitted in a consultation and an AI-written summary of them, how can we measure which groups the summary underrepresents, and test the measure on real data?

Past and Current Research

A geometry of collective decisions. How similar are two elections? A useful distance between elections must ignore what the candidates are called and in which order the voters are listed, since two elections can have the same structure with entirely different candidates (presidential candidates vs. types of sushi). We showed that the most natural such distances, and the related problem of finding one election inside another, are computationally hard [C2, J3, C5, J2]. I proposed instead the *positionwise distance*, which compares two elections by how often each candidate is ranked in each position, after optimally matching the candidates of the two elections. It can be computed efficiently, and it made it possible to draw the *map of elections* [C3]: compute the distances between all pairs in a collection of elections and draw them in the plane, so that similar elections lie close together. We then added landmark elections that make its regions

interpretable [C4]; the complete framework appeared in *Artificial Intelligence* [J4].

Maps also show which parts of the space an experiment covers and where real elections lie relative to synthetic ones. For example, the map showed that real-life elections occupy only a small region of the space and that elections from the Mallows model, the most common model of noisy rankings, pass through it, so suitable Mallows parameters generate realistic synthetic data. It also showed that the standard Mallows model drifts toward identical votes as the number of candidates grows, which led us to normalize its parameter [C4], and that integer programming solvers for some multiwinner rules take longest on elections close to impartial culture [T1]. The same approach works beyond ordinal elections. We built maps of approval elections [C7], of elections of different sizes [C19], of stable roommates and stable matching instances [C9, J1] (Best Student Paper Award), of tournaments [C22], and of fair division instances [C25], and used the geometry for preference learning [C8]. To help others use this methodology, I co-authored a guide to numerical experiments on elections [C15] and developed two Python libraries: `mapof`, for drawing maps, and `prefsampling` (with Simon Rey), for sampling synthetic preferences.

I also measure the structure within a single election, and these measures are the starting point for the auditing tools I propose below. We introduced indices of agreement, diversity, and polarization based on *k-Kemeny*, which seeks k representative rankings instead of one consensus [C10, C24, C26]. We also look for hidden structure in preference data, such as consistent subelections [C13] or the most conflicting pair of candidates [C18]; finding the groups of voters that an outcome serves badly is a question of the same kind.

Real data and practice. In participatory budgeting (PB), residents vote on how to spend part of a city's budget. I founded Pabulib in 2020 and maintain it [C11]. Our later work on PB includes learning realistic models of approval voting [C20], measuring how well individual projects performed under a given rule [C17], and a game-theoretic analysis of how proposers set project costs under the Method of Equal Shares [C23]. Under this rule (Peters, Pierczyński, and Skowron, 2021), each resident receives an equal share of the budget, and a project is funded only if its supporters can jointly pay for it from their shares, so that each group of residents gets a fair part of the budget. The team I belong to convinced Polish cities, including Wieliczka and Świecie, to adopt it.

Toward language. At the University of Geneva, I study collective decisions in international politics, combining voting records with NLP pipelines that extract structured information from large collections of diplomatic and policy texts.

Simulating democratic deliberation with LLM agents. I am a coauthor of *VoteSim* [C28], developed with collaborators at the University of Toronto and Google DeepMind, which simulates a full democratic cycle with LLM voters based on real participants' profiles. Across seven language models, the voters preferred the bills of parliaments elected under proportional systems to those of majoritarian ones.

What an abstention means in the UN General Assembly. Abstention is the most common way states withhold support in the General Assembly (11.8% of roughly 860,000 votes in 1946–2025), yet standard measures of state preferences treat every abstention alike. In ongoing work that I lead [P2], we classify all 101,070 abstentions by the context in which they were cast, and show that this context tracks international conflict while the overall abstention rate does not. We label each resolution's subject with a language model, which makes the record a large collection of votes over texts.

Research Agenda

Social choice models treat the alternatives as abstract symbols, whereas a city project or a UN resolution is a text with content. Fish et al. (2024), under the name *generative social choice*, prove proportionality guarantees for slates of LLM-generated statements, assuming that the LLM answers its queries accurately, and Tessler et al. (2024, *Science*) found that, in groups of five,

statements by an LLM mediator (the *Habermas Machine*) were accepted more often than those of human mediators. In a position paper from a Dagstuhl seminar, we chart how text can enter every stage of collective decision making [P1]. My agenda focuses on evaluating text-based outcomes on real data, and it continues each of the three lines of work above in one research direction.

Direction A: A geometry of collective decision problems, from ballots to text. Here I want to extend maps of elections to texts and to much larger instances.

- **Maps of deliberations.** When alternatives are texts, two decision problems can be similar in how people voted, in what the alternatives say, or both. One approach is to represent each alternative both by its votes and by an embedding of its text, and to match the alternatives of two instances with a cost that combines the two. With such distances, we could compare human and AI deliberations with each other and place sessions of the UN General Assembly on one map; they would also be the basis for comparing text-based outcomes in Direction C.
- **Computing invariant distances at scale.** The natural isomorphism-invariant distances are hard to compute, and the positionwise distance is an efficient substitute for ordinal elections. Its cost, however, grows quickly with the number of candidates, and a map needs the distance between every pair of elections. How can maps be made feasible for elections with many candidates, such as large participatory budgets, and for very large collections of elections? For which other kinds of instances do efficient substitutes exist that give nearly the same maps as the exact distances?
- **Benchmarks with provable coverage.** How many instances, and which ones, must a benchmark contain to cover the space of instances up to a given precision under a given distance? Which samplers achieve this? The answers will become defaults in `prefsampling` and `mapof`.

Direction B: Participatory democracy as a living laboratory. Participatory budgets provide real ballots over projects that come with text descriptions (which Pabulib does not yet store), which makes them a good source of data for this agenda and a setting in which predictions about proportional rules can be tested.

- **Votes over texts.** I will add the project descriptions, residents' comments, and outcomes over several years to Pabulib, turning it into a longitudinal benchmark in which every ballot refers to a text. The UN General Assembly record, with 5,693 resolutions adopted by recorded vote together with their full texts [P2], is another source of votes over texts, from a very different setting. Both datasets will also be used in Direction C.
- **Does proportionality carry over to content?** Proportional rules for participatory budgeting have been observed to spread funds evenly across project categories [C11] (Nelissen, 2023). With project texts, we can ask whether they also give each group of residents a proportional share of the *topics* it cares about, and design rules that guarantee it.
- **Does the rule change the process?** We have analyzed how proposers choose project costs under the Method of Equal Shares [C23]. Field studies with partner cities can test whether the rule also changes who submits projects and who votes.

I will seek partnerships with Swiss municipalities for the field studies, building on my experience of working with Polish cities on the Method of Equal Shares. Until such partnerships are in place, the studies can start with Polish cities that use the Method of Equal Shares and with existing Pabulib data.

Direction C: Representation when opinions are expressed in text and aggregated by AI. This direction is the core of the agenda. It asks how to check whether a text that AI produces from many opinions represents those opinions fairly.

- **Auditing summaries and consensus statements.** Given opinions written in natural language and a summary or consensus statement, how should we measure how proportionally it represents the population? Which measures satisfy natural axioms, and which can be estimated with few LLM queries and provable error bounds? One starting point is to treat the summary as a set of statements and to estimate, with LLM queries, how strongly each

participant would endorse each of them, as generative social choice does while constructing a slate. The difference is that here the text is given, produced by any method, and the question is how well it represents the participants. Proportionality axioms for committees then become measurable properties of the text, which real consultation data can test. Because the endorsements are themselves estimated by a language model, the measures must be validated against participants' own judgments wherever these are available.

- **An LLM aggregator is a voting rule.** Fix a language model and a prompt that turns a set of opinions into k statements, and we obtain a multi-winner voting rule whose input is text. Existing methods, such as generative social choice and LLM mediators like the Habermas Machine, can then be tested like any other rule: does the outcome change when the same opinions are presented in a different order (anonymity)? Can a participant steer it by writing a manipulative opinion (manipulability)? Does it cover each group in proportion to its size? I helped establish this kind of large-scale numerical experiment for classical rules [C15]. The result will be a public test suite that any proposed aggregator can be run against.
- **Filling in sparse votes with text.** In online consultations, each participant reacts to only a small fraction of the comments; on the consultation platform Pol.is, about 9% [P1]. Language models can predict the missing reactions from the texts, but their errors then propagate to the outcome. How large is this effect, and which aggregation rules are robust to it?

Goal for the first five years. By the end of this period, my group will have produced maps of deliberations (A), a public benchmark of votes over texts built from participatory budgets and the UN General Assembly record (B), and open, validated tools for auditing AI-written summaries and consensus statements (C).

Students, funding, and service. Possible first PhD topics are scalable invariant distances (A, theoretical), proportional coverage of topics in participatory budgeting (B, applied and data-centric), and testing LLM aggregators as voting rules (C, empirical and NLP-oriented). I will apply to the Swiss National Science Foundation early on and for an ERC Starting Grant built around Direction C. I serve on the editorial board of JAIR (since 2026) and on the program committees of NeurIPS, AAAI, AAMAS, and IJCAI.

Why EPFL

Computational social choice would be a new area at IC, and my agenda shares questions with several colleagues:

- *Bryan Ford* works on proof of personhood and e-voting, the infrastructure that online democratic processes need. Online participatory budgeting needs both his work, to count each resident once while keeping votes secret, and mine, to choose and test the rule that aggregates the votes.
- *Robert West*'s lab studies how language models shape online discourse, for example how persuasive they are in debates. Direction C asks the complementary question: whose opinions does an AI system represent when it aggregates them?
- *Antoine Bosselut*'s group builds and evaluates open language models, including Apertus. Checking whether a model represents the opinions it summarizes would be a new kind of evaluation, and an open model makes such audits reproducible.

I also want to do this work in Switzerland because of its strong tradition of voting. Swiss citizens vote on public questions several times a year, and in 2023 the city of Aarau used the Method of Equal Shares for its participatory budget. Geneva, where I already work, hosts the UN Human Rights Council, whose votes could extend my work on the General Assembly [P2].

Other references. Fish, Gölz, Parkes, Procaccia, Rusak, Shapira, and Wüthrich. Generative social choice. *Proceedings of EC-2024*. Nelissen. An empirical analysis of participatory budgeting in Amsterdam. arXiv:2310.18033, 2023. Peters, Pierczyński, and Skowron. Proportional participatory budgeting with additive utilities. *Proceedings of NeurIPS-2021*. Tessler, Bakker, Jarrett, et al. AI can help humans find common ground in democratic deliberation. *Science* 386(6719), 2024.